<u>Executive Summary</u>

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This report presents a full- cycle quantitative research study on momentum, reversal and statistical arbitrage across 25 cryptocurrencies 2021-2026 Fourteen momentum signals and five reversal based signals were tested using dollar neutral long-short framework with daily rebalancing and 20 bps one-way transaction cost assumption.

The sample results return 30 day volume weighted momentum as the strongest individual signal. However, the strategy's Sharpe Ratio declined from 1.95 during a training period in 2021-2022 during a training period to 0.14 during the 2023-2025 validation period; an indication of performance decay and limited evidence of stable out-of-sample alpha.

The most credible result came from the quarterly walk-forward portfolio test. The inverse-volatility achieved a 31.85% annualised return, a Sharpe ratio of 0.64, a maximum drawdown of -62.59% and a 50.8% win rate.

The report supports diversified, cost-aware momentum portfolio construction rather than reliance on a single standalone signal, although the substantial train-to -validation Sharpe decay means that the strategy should be treated as an exploratory research.

1. Research Design and Methodology

1.1 Universe Data

The study universe comprised 25 cryptocurrency spot pairs traded against USDT on Binance. Daily OHLCV data was acquired from Binance public endpoints using the CCXT library. The daily dataset cover the period from 1 January 2021 to 27 June 2026, providing 2,004 daily observations.

The universe included majors asset such as BTC, ETH, BNB, SOL, XRP, DODGE, ADA AVAX, LINK and other large-cap cryptocurrencies. Assets with fewer than 200 days of available history are excluded from the final daily analysis to reduce the impact of insufficient price histories. The benchmark being BTCUSDT buy and hold. The sample had a training period from 2021-2022 and a separate validation period from 2023-2025, so the strategy can be assessed out of sample.

1.2 Signal Universe

Nineteen signals constructed across the two strategy families:

- Momentum: 14 variants. Time series momentum using 1-, 3-, 7-, 14-, 30-, 60- and 90 day lookback windows; cross-sectional momentum suing 7-, 30- and 90 day windows; volume-weighted momentum7-, 14- and 30 days windows; and a weekday- seasonality momentum signal.

- Reversal and Statistical Arbitrage: 5 Variants. These included 1-day and 3 day short- horizon reversal, low volume conditional reversal, pairs-spread reversal and BTC volatility conditioned macro reversal.

The momentum signals were based on historical log returns, cross-sectional ranks or relative trading volume. The pairs strategy estimated rolling spread z-scores between sufficiently correlated cryptocurrency pairs, while the macro-regime strategy adjusted reversal exposure according to BTC realised volatility.

1.3 Backtesting Framework

All strategies were evaluated within an unconstrained long-short framework. At each daily rebalance, the strategy went long the top quintile of assets ranked by the relevant signal and short the bottom quintile. Long and short positions were independently normalised so the long leg summed to +1.0 and approx. zero net market exposure.

The backtester applied a one-period lag between signal formation and portfolio returns. Information available through the close of day t was used to determine weights for the return on day t+1 intended to prevent look-ahead bias.

2. Results Overview

2.1 Strategy Ranking By Net Sharpe Ratio

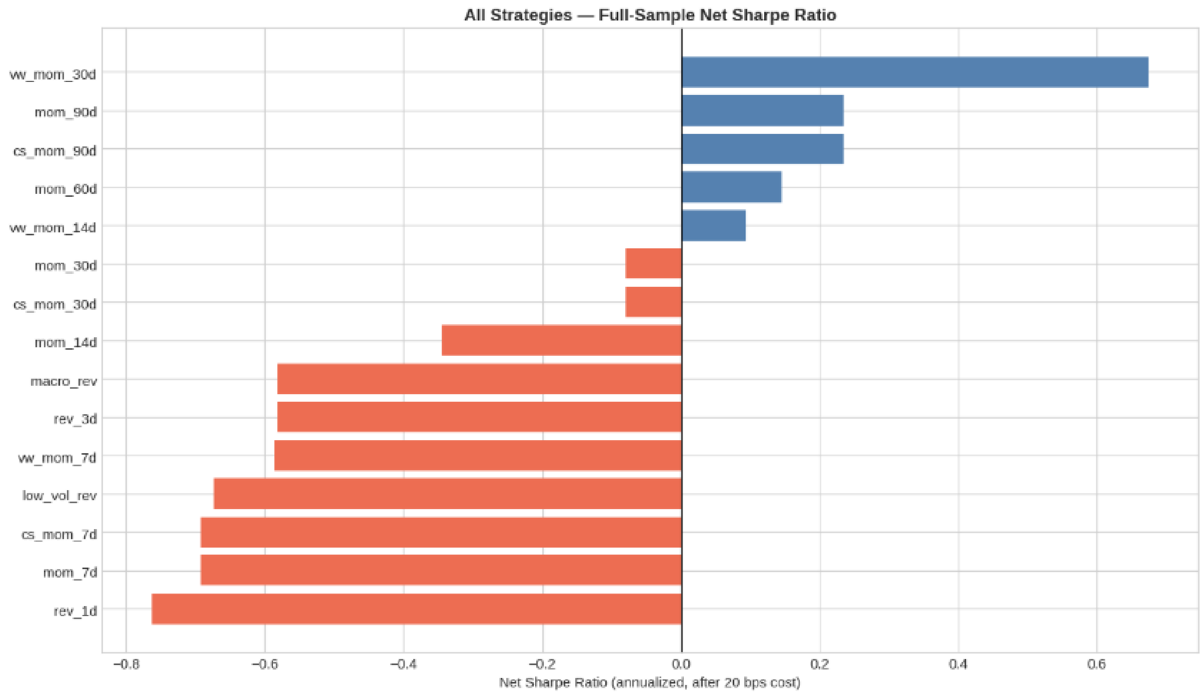

Figure 1 All strategies ranked by annualised net Sharpe ratio. Blue = Positive, Red = Negative

Tier	Strategies	Characterisation
Tier 1 — Positive full-sample momentum	vwmom30d, mom90d, csmom90d, mom60d, vwmom14d	Sharpe 0.13–0.74; positive returns, but material validation decay
Tier 2 — Marginal or weak	mom30d, csmom30d, mom14d, vwmom7d, mom7d, csmom7d, mom1d, mom3d, weekdaymom	Sharpe N/A to -1.41; costs and drawdowns remove or reverse the apparent edge
Tier 3 — Reversal/statistical-arbitrage destroyers	rev1d, rev3d, lowvolrev, pairsrev, macrorev	Sharpe -0.57 to -0.88; maximum drawdowns approximately -95.9% to -99.9%; structurally loss-making

3. Full Strategy

ALL STRATEGIES – full sample – ranked by net Sharpe (after 20 bps cost)

Strategy	Ann. Return (net) %	Ann. Volatility %	Sharpe Ratio	Max Drawdown %	Alpha (ann) %	Alpha t-stat	Sig	Beta vs BTC	Win Rate %	Avg Daily Turnover
vw_mom_30d	40.410	59.950	0.670	-67.250	68.120	2.060	**	-0.030	48.860	0.250
mom_90d	13.320	56.790	0.230	-70.170	35.800	1.280	ns	-0.060	50.660	0.190
cs_mom_90d	13.320	56.790	0.230	-70.170	35.800	1.280	ns	-0.060	50.660	0.190
mom_60d	8.600	59.190	0.150	-74.610	31.160	1.090	ns	-0.050	49.780	0.240
vw_mom_14d	5.720	61.090	0.090	-68.710	27.310	0.940	ns	-0.010	47.890	0.390
mom_30d	-5.350	67.310	-0.080	-84.120	20.380	0.660	ns	-0.080	48.660	0.330
cs_mom_30d	-5.350	67.310	-0.080	-84.120	20.380	0.660	ns	-0.080	48.660	0.330
mom_14d	-23.260	67.450	-0.340	-89.590	-2.250	-0.080	ns	-0.070	47.010	0.490
macro_rev	-57.470	98.730	-0.580	-99.740	-34.820	-1.030	ns	0.170	51.530	0.970
rev_3d	-57.470	98.730	-0.580	-99.740	-34.820	-1.030	ns	0.170	51.530	0.970
vw_mom_7d	-38.480	65.660	-0.590	-96.560	-23.190	-0.950	ns	-0.040	45.410	0.590
low_vol_rev	-60.330	89.660	-0.670	-99.850	-49.600	-1.840	*	0.280	43.950	1.140
cs_mom_7d	-47.920	69.280	-0.690	-98.950	-32.030	-1.330	ns	-0.100	45.600	0.670
mom_7d	-47.920	69.280	-0.690	-98.950	-32.030	-1.330	ns	-0.100	45.600	0.670
rev_1d	-59.580	78.130	-0.760	-99.710	-42.440	-1.680	*	0.050	53.040	1.590
pairs_rev	-38.530	43.790	-0.880	-95.900	-33.010	-2.170	**	0.030	48.710	0.290
mom_1d	-88.240	73.670	-1.200	-100.000	-84.460	-5.980	***	-0.040	37.580	1.590
weekday_mom	-66.040	47.220	-1.400	-99.800	-61.980	-4.850	***	0.020	43.560	1.290
mom_3d	NaN	111.760	NaN	-99.890	-67.950	-2.420	**	-0.200	43.270	0.970

Table 1 Full Performance Metrics(After 20 bps cost)

3.1 vm_mom_30d

vm_mom_30d is the strongest individual strategy in the fall sample results, gaining 40.41% annualised net return and a net Sharpe ratio of 0.67 after 20 bps transation costs. It produced annualised alpha 68.12% relative to BTC buy-and-hold and beta of -0.03. Its average daily turnover of 0.25 is relatively moderate compared with the higher turnover short horizon strats.

3.2 Volume weighting effect

The results provide evidence in favour of volume weighting at the 14-day horizon. Vw_mom_14 generating a 5.72% annualised return and a Sharpe ratio of 0.09, compared with -23.26% annualised return and a -0,24 sharpe ratio for mom_14d. The volume weighted strategy also experience smaller maximum drawdown, at 68.71% compared with -89.59% .

3.3 Why reversal strategies failed

All five reversal and statistical-arbitrage strategies produced negative risk-adjusted performance after the assumed 20bps one way transaction cost. Their Sharpe ratios ranged from -0.58 to 0.88, while annualised net returns ranged from -38.53 to 60.33%. Maximum drawdowns were severe, ranging from -95.90% for pairs_rev to -99.85% for low_vol_rev. Average daily turnover varied from 0.29 for pais_rev to 1.59 for rev_1d, increasing the sensitivity of shorter horizon signals to execution costs.

4. Equity Curves & Drawdown Analysis

4.1 Top – 5 Strategy Equity curves

Table 2 Cumulative portfolio value for five strategies with the highest validation. Return net of 20 bps one way transaction cost

The equity curves compares the five strategies ranked most highly by validation -period Sharpe ratio with BTC buy-and-hold. The strategies show significant variation in cumulative wealth and drawdown behaviour, although their performance paths remain volatile and show periods of substantial divergence.

- Training Period, 2021 – 2022: Strategy performance varied considerably across signals, with several strategies experiencing large drawdowns during the 2022 crypto-market downturn.
- Validation period 2023 – 2025: The curves provide a more demanding test of generalisability. Performance became more differentiated, but the validation results were materially weaker than the training results for most strategies, indicating Sharpe decay and possible overfitting .

4.2 Drawdown Profile

Figure 3 Rolling Drawdown (%) for Top 5 Strategies

Among the principal momentum strategies, maximum drawdowns ranged from approx. -67.25% for vmmom30d to -87.71% for mom14d. The weaker short horizon momentum and reversal strategies experienced considerably more serve losses, with maximum drawdowns between -95.58% and -100.00%

This should be considered with annualised volatility: vwmom30d recorded annualised volatility of 60.50%. mom90d and csmom90d recorded 57.36%.

The drawdown curves also show that losses were not confined to a single strategy. Multiple strategies experienced drawdowns during similar periods, indicating the presence of common market wide shocks and correlated risk factors.

5. Rolling Sharpe Analysis

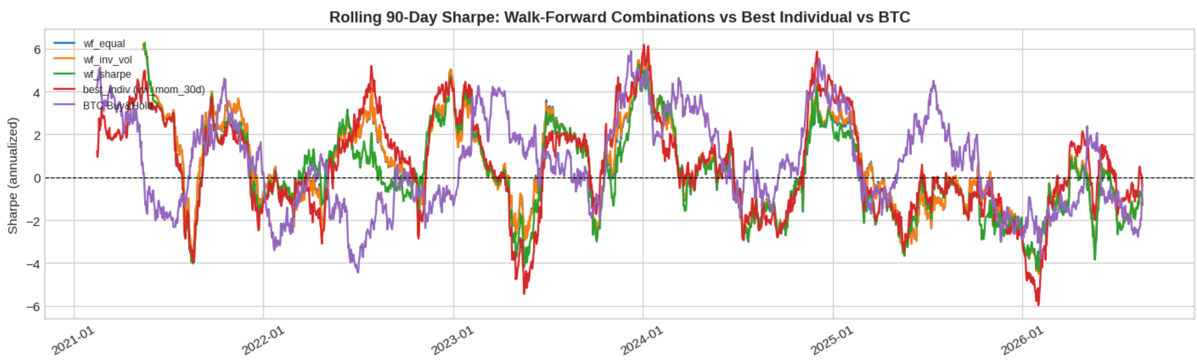

Figure 4 90 Day rolling annualised Sharpe Ratio for top strategies and combined portfolio

The rolling 90 day Sharpe analysis compares the out-of-sample walk forward combination portfolios with the strongest individual strategy and the BTC benchmark.

Individual strategy exhibits greater variation in rolling risk-adjusted performance, reflecting sensitivity to changing market conditions. The equal weighted and inverse volatility portfolios display smoother performance patterns because they diversify exposure across multiple signals. The inverse-volatility portfolio produced the strongest walk forward result with a Sharpe ratio of 0.64, narrowly exceeding the equal weighted portfolio's 0.63.

The sharpe -weighed portfolio achieved a lower Sharpe ratio of 0.41, therefore historical Sharpe performance is not a reliable basis for allocating future portfolio weights.

6. Combined Portfolio Analysis

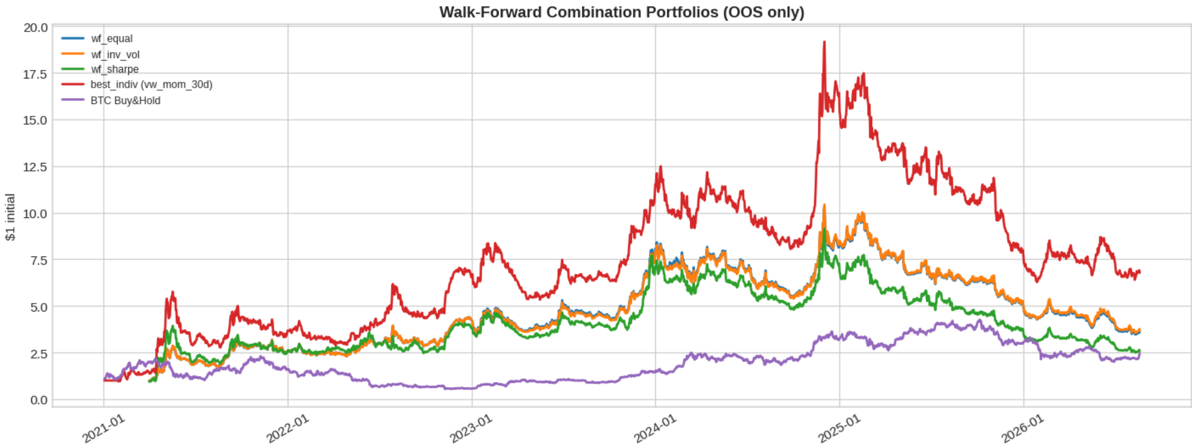

Figure 5: Combined portfolio equity curves vs best individual strategy

Portfolio	Strategy mix	Key benefit
Equal Weight	vwmom30d + mom60d + mom90d + csmom90d	Simple diversification across signals with positive validation Sharpe
Inverse-Volatility Weighted	vwmom30d, mom60d, mom90d and csmom90d, weighted by inverse 60-day volatility	Larger allocation to lower-volatility strategies; aims to smooth portfolio risk
Sharpe Weighted	Same positive-validation strategy set, weighted by trailing 252-day positive Sharpe	Tilts toward strategies with stronger recent risk-adjusted performance

Observations from Figure 5

Figure 5 compares three quarterly walk-forward combination approaches using strategies with positive validation-period Sharpe ratios. The equal weighted portfolio assigns the same allocation to each strategy and a simple diversification benchmark.

The inverse volatility portfolio uses trailing 60 day volatility to allocate more weight to lower- volatility strategies, while the Sharpe- weighted portfolio uses trailing 252 day positive Sharpe ratios to favour strategies with stronger risk adjusted performance.

7. Transaction Cost Sensitivity

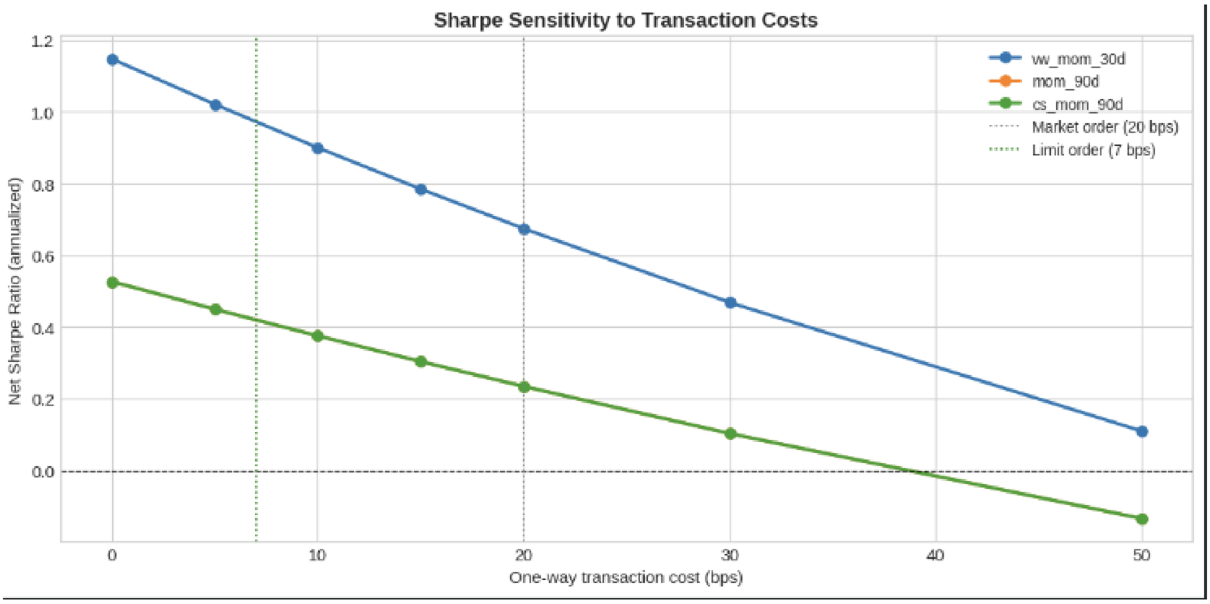

Figure 6: Sharpe Ratio sensitivity to transaction costs for top 2 strategies

Figure 6 shows that transaction costs have a material effect on the apparent profitability of the strategies. Vmom30d is the most cost resilient strategy in the full sample analysis, remaining above a zero Sharpe ratio until approximately 30 bps. The validation analysis shows that vmom30d;s Sharpe ratio fell from 1.95 during the training period to 0.14 during validation.

- Limit order scenario: At the 7bps limit order assumption, both strategies profitable and retail relatively strong Sharpe ratios, highlighting the importance of execution quality in a high-volatility, high turnover asset class
- Break-even Costs: Estimated Break even level is approximately 50 bps for vm303d, while mom90d remain profitable up to approximately 30bps per one way
- Operational Implication: Reducing execution costs from the 20 bps market order assumption toward the 7 bps limit order assumption should improve performance. To be consider, however, the cost sensitivity analysis is based on a flat cost model and does not account for liquidity variation, bid-ask spread widening or the practical cost of shorting cryptocurrency spot pairs.

8. Recommendations

8.1 Walk Forward Diversified Momentum Portfolio

Based on the out-of-sample portfolio tests, the preferred approach is a quarterly walk-forward diversified portfolio using inverse -volatility weighting. Unlike a fixed allocation, the walk forward methodology estimates portfolio weights using only information available before each subsequent quarter reducing hindsight bias and provides a more realistic assessment of how a portfolio would perform in implementation.

The inverse volatility approach is preferred because it delivered the strongest walk-forward result, with a 31.85% annualised return, a Sharpe ratio of 0.64 and a maximum drawdown of -62.59%. However, the advantage over the equal-weighted portfolio was small. The equal-weighted portfolio generates a 31.23% annualised return.

Portfolio approach	Strategy selection	Weighting method	Recommendation	Rationale
Walk-forward inverse-volatility	Strategies with positive validation Sharpe ratios	Weights proportional to inverse trailing 60-day volatility	Preferred approach	Highest walk-forward Sharpe ratio and lowest maximum drawdown among tested combinations
Walk-forward equal weight	Strategies with positive validation Sharpe ratios	Equal allocation to each eligible strategy	Recommended simpler alternative	Almost identical performance with lower implementation complexity
Walk-forward Sharpe weighting	Strategies with positive validation Sharpe ratios	Weights proportional to positive trailing 252-day Sharpe ratios	Not preferred	Lower Sharpe ratio and larger maximum drawdown

Table 3 Portfolio- Construction Recommendation Note: Portfolio weights are recalculated quarterly using information available up to the relevant quarter end and then applied to the following quarter. Returns are net of the 20 bps one-way transaction cost assumption

8.2 Portfolio Performance

Table 3 summaries the out of sample results for the portfolio construction methods. Producing the highest annualised return and Sharpe ratio was the inverse volatility portfolio with the smallest maximum drawdown. In the contrary, the Sharpe-weighted portfolio produced weaker results, suggesting that allocating capital according to recent historical Sharpe doesn’t improve subsequent portfolio performance.

Portfolio	Annualised Return	Sharpe Ratio	Maximum Drawdown	Win Rate	Interpretation
Equal-weighted	31.23%	0.63	-63.05%	50.6%	Strong simple benchmark

Portfolio	Annualised Return	Sharpe Ratio	Maximum Drawdown	Win Rate	Interpretation
Inverse-volatility weighted	31.85%	0.64	-62.59%	50.8%	Preferred portfolio method
Sharpe-weighted	23.30%	0.41	-69.82%	49.3%	Weaker risk-adjusted performance

8.3 Implementation controls and Risk Controls

- Think about the implementation as a research exercise, not as a live-trading strategy that is validated.

- Consider the potential weaknesses of the historical backtests, including transaction costs assumptions, survivorship bias, liquidity considerations and challenges with maintaining shorts in crypto.

- Assume that strategy performance is regime-specific. Each signal becomes very weak when the market environment is different from the one in which it used to perform well.

- Approach the full sample results of vwmom30d with caution. Despite having a good Sharpe ratio of 0.74 and an annualised net return of 44.92%, vwmom30d's Sharpe ratio fell from 1.95 during the 2021–2022 training period to 0.14 during the 2023–2025 validation period.

- Interpret the 92.8% decrease in the vwmom30d Sharpe ratio as an indication of unreliability of full-sample performance in forecasting future returns.

- Recalculate walk-forward portfolio weights quarterly using historical data known at the end of the previous quarter.

- Calculate realised transaction costs (commissions, spreads, slippage and market impact), instead of assuming 20 bps transaction costs.

- Take 20 bps per one-way trade as a base-case market-order assumption.

- bps sensitivity-case assumption.

- Start with lower notional amount of the trade until the realised turnover, slippage, net returns and drawdowns align with the back tested model.

- Compare live net performance with BTC buy-and-hold and the walk-forward benchmark.

- Use position, loss and portfolio-risk limits since dollar neutrality does not mean lack of drawdown, liquidity or factor risk.

- Monitor realised portfolio drawdowns in comparison with the historical walk-forward maximum drawdown of -62.59% of the inverse volatility portfolio and -63.05% of the equal weight portfolio.

- Reduce exposure or stop trading if the realised transaction costs exceed the cost tolerance in the model, turnover increases significantly, and net performance is persistent worse than the walk-forward benchmark.

- Do not expand the capital allocation because of strong performance of a strategy in a short period of time; the improvement should be sustainable over several rebalancing periods.
- Examine if limit order execution allows reducing the realised costs closer to 7

8,4 What to avoid

- Do not run rev1d, rev3d, lowvolrev, pairsrev or macrorev on their own as strategies.

- Do not run mom1d, mom3d, mom7d, csmom7d, vwmom7d or weekdaymom on their own as live strategies.

- Do not make the best full sample strategy into the automatic live recommendation.

- Do not rely on Sharpe-weighted allocation for building the portfolio.

- Do not interpret BTC beta being close to zero to mean the portfolio is safe.

- Do not overlook the issues of shorting, liquidity, borrow, funding and execution limitations on exchanges.

9. Conclusion

The study demonstrated that diversified cost aware momentum portfolios produce positive historical risk adjusted return in crypto markets, although strength of individual signal declines out of sample, based on 25 cryptocurrency markets, daily rebalancing and a 20 bps one way transaction assumption.

Core Findings:

- Full sample momentum performance is strongest for volume weighted signals.
- Reversal does not work at 20 bps
- Walkforward combination improves portfolio robustness
- Combination adds risk reduction rather than guaranteeing higher returns
- Execution quality remains important
- The strategy should be treated as a research candidate rather than a validated live system